TESTING OF HYPOTHESIS

Introduction :

In most of the situations it is necessary to make decisions or to test the assumption regarding the parameter of the population rather than estimating them.

For example, if two fertilizers are to be tested with respect to the yields, we have to frame a statement about the fertilizers and we have to examine whether the formed statement is correct or not. Similarly it is necessary to check the average income of an employee of a particular area exceeds the predetermined value, we have to frame the statistical statement about it. Such statements about the population parameter are called **statistical hypothesis** and there's testing procedure can be called as **testing of hypothesis**.

Statistical hypothesis :

The statement formulated on statistical population are called statistical hypothesis. It means a statistical hypothesis reveals the probability distribution of the population or the parameter involved in it.

Suppose we are interested to examine two fertilizers A and B, according to their yields. In this contest we can formulate the following statements about the fertilizers A and B.

1. The fertilizer A provides more yield than the fertilizer B
2. The fertilizer B provides more yield than the fertilizer A
3. There is no significant difference between the two types of fertilizers A and B with respect to their yields.

The above three statements are called statistical hypothesis. The third statement is an unbiased statement, where as the remaining two statements shows Bias towards either of the fertilizers.

Null hypothesis :

The Statistical hypothesis which is to be tested is called **null hypothesis**. It is denoted by H_0.

Null hypothesis should be numerical or natural regarding to the outcome of the test.

It should be completely impartial and should not allow any personal views to influence the decision.

For example, let us suppose we have two educational systems A and B and we have to examine which system is better, then in this contest the null hypothesis will be framed as H_0 : There is no significant difference between the two educational systems.

Alternate or alternative hypothesis :

The decisions which we are accept when the null hypothesis is rejected is called **alternative** hypothesis.

Statistical hypothesis or statement which is made exact opposite to the null hypothesis is called an **alternative** hypothesis. It is denoted by H_1.

Example: If the null hypothesis, in testing of unbiasedness is $H_0 : p = \frac{1}{2}$ then the alternative hypothesis is $H_1 : p \neq \frac{1}{2}$ (or) $H_1 : p > \frac{1}{2}$ (or) $H_1 : p < \frac{1}{2}$

Simple and composite hypothesis :

Simple hypothesis :

A Hypothesis which can specify all the parameters of the population completely is known as simple hypothesis.

Example : Suppose we have a sample of 'n' observations, with the values $x_1, x_2, x_3, ...x_n$ are drawn from a normal population with mean μ and variance σ^2 then $H_0 : \mu = \mu_0$ and $H_0 : \sigma^2 = \sigma_0{}^2$ is known as simple hypothesis.

composite hypothesis :

A statistical Hypothesis which does not specify all the parameters of the population completely is known as composite hypothesis.

Example : $H_1 : \mu > \mu_0$ (or) $H_1 : \mu < \mu_0$

$H_1 : \sigma^2 > \sigma_0{}^2$ (or) $H_1 : \sigma^2 < \sigma_0{}^2$

Two types of errors :

Error may happen while decisions are being made with the help of the sample instead of considering the total population in testing of hypothesis. In such cases the conclusions drawn (accepting or rejecting null hypothesis) on the basis of the sample data may not be true in all the cases. So we may commit two types of errors which are exhibited as follows

	H_0 **is True**	H_0 **is False**
Rejection of H_0	**Type-I Error**	**No Error**
Acceptance of H_0	**No Error**	**Type-II Error**

Type –I Error :

Probability of rejection of null hypothesis when it is true is called Type-I error.

(or)

Probability of accepting the alternative hypothesis when it is false is known as Type-I error.

Type-I error is also known as producer's risk and it is denoted by 'α'.

$$Type-I\ Error(\alpha) = P\left[\text{ Rejection of } \mathrm{H}_0 / H_0\ is\ True\right]$$
$$= P\left[Accepting\, \mathrm{H}_1 / H_1 \text{ is False}\right]$$
$$= P\left[x \in w / H_0\right]$$
$$\alpha = \int_w L_0\ dx$$

Type-II Error :

Probability of accepting the null hypothesis when it is false is called Type-II error.

(or)

Probability of rejecting the alternative hypothesis when it is true is known as Type-II error.

Type-II error is also known as consumer's risk and it is denoted by 'β'.

$$Type-II\ Error(\beta) = P\left[\text{ Accepting of } \mathrm{H}_0 / H_0\ is\ False\right]$$
$$= P\left[Rejecting\, \mathrm{H}_1 / H_1 \text{ is True}\right]$$
$$= P\left[x \in \overline{w} / H_1\right]$$
$$\beta = \int_{\overline{w}} L_1\ dx$$

Critical Region :

Critical region is also known as rejection region.

Let $x_1, x_2, x_3, \ldots\ldots\ldots x_n$ be the 'n' sample observations.

Let us consider these 'n' sample points in a sample space.

Let us check whether the sample points are lie within the region (or) outside the region.

The total sample space S can be divided into two disjoint sets $w\, and\ (S-w)\overline{w}$

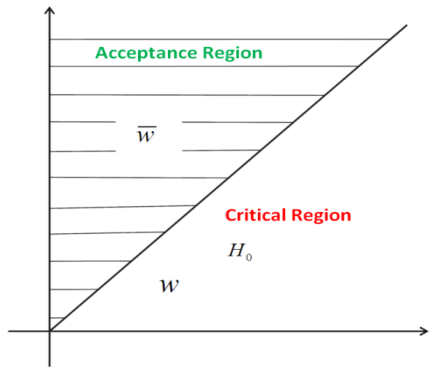
The null hypothesis H_0 is to be reject if the observed sample points falls in w and if they falls in $\overline{w}$ then we can reject H_1 and accept the null hypothesis H_0.

The region of rejection of null hypothesis when it is true is the region of outside of the set where H_0 is rejected. If the sample points falls in that region then it is called critical region.

Critical region is the region of the sample space when H_0 rejected even though it is true. When the sample statistic falls in that region then $S = w \cup \overline{w}$ and $w \cap \overline{w} = \phi$.

A Region in which we can reject a null hypothesis at a certain probability is called critical region.

In other words critical region is a region such that an observation falls in it tends to rejection of null hypothesis when it is true.



Tests of significance :

In testing of hypothesis a hypothesis is to be framed about the population parameter and then we have to test whether the framed hypothesis is correct or not. If the formed null hypothesis refer the parameters completely then it is called null hypothesis.

After framing the null hypothesis our business is to take the decision about the rejection or acceptance of the null hypothesis. The methods which are used for this purpose are known as tests of significance.

Example : Let us suppose that, the bulbs of a company are manufactured by using two methods old method and new method.

Here we have to decide which method increases the lifetime of the bulbs, in this case we can frame 3 statements. they are

- Old method is better than new method.
- New method is better than old method.
- There is no significant difference between the old method and the new method.

We have to decide which of the above statements are true, to make a decision about the method of manufacturing. Here we have to use a systematic procedure for taking decision about the statements, the procedure which is used is known as the tests of significance.

Power of the Test :

$1-\beta$, the probability of rejection of the null hypothesis H_0 when it is false is known as power of the test.

Thus,

$$\beta = P\left[\text{Accepting of } \mathrm{H}_0 / H_0 \text{ is False}\right]$$
$$= P\left[\text{Rejecting } \mathrm{H}_1 / H_1 \text{ is True}\right]$$
$$= P\left[x \in \overline{w} / H_1\right]$$
$$\beta = \int_{\overline{w}} L_1 \, dx$$
$$1-\beta = 1-\int_{\overline{w}} L_1 \, dx$$

LARGE SAMPLE TESTS

Test for the significance of single sample mean($\bar{x}$) :

Let $x_1, x_2, x_3, \ldots\ldots x_n$ be a random sample drawn from the normal population with mean μ and variance σ^2 for the given data of sample observations we can calculate the sample mean $\bar{x}$ and
variance s^2 as follows .

$$\bar{x} = \frac{\sum x_i}{n} \quad \text{and} \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Case I : For testing a single mean when the population variance σ^2 is known :

For testing a single mean we have to formulate the null hypothesis as

H_0 : There is no significant difference between the sample and population means .

$H_0 : \mu = \bar{x}$

Since the sample is large, we have to use the Z – Statistic to test about the null hypothesis as follows:

Here $t = \bar{x}$ Standard Error of x is $\frac{\sigma}{\sqrt{n}}$

$\therefore$ The test statistic $Z = \frac{t - E(t)}{S.E(t)}$

$$Z = \frac{\bar{x} - E(\bar{x})}{SE(\bar{x})}$$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \approx N(0,1)$$

Case II : For testing a single mean when the population variance σ^2 is not known :

If the population variance σ^2 is not known then we can test the significance of means by using the following test statistic .

$$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}} \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= $\text{Statistic} \pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t) = \bar{x} \pm Z_{\frac{\alpha}{2}} \left(\sigma / \sqrt{n} \right)$$

Test for the significant difference between the two means :

Let $x_{11}, x_{12}, x_{13}, \ldots\ldots x_{1n_1}$ and $x_{21}, x_{22}, x_{23}, \ldots\ldots x_{2n_2}$ be the values of two random samples of sizes n_1 and n_2 drawn from the two different populations with the parameters μ_1, μ_2 and σ_1^2, σ_2^2.

i.e., $X_1 \approx N(0,1)$ and $X_2 \approx N(0,1)$

From the given two samples

Mean of the first sample $\bar{x}_1 = \dfrac{\sum_{i=1}^{n_1} x_{1i}}{n_1}$

Mean of the Second sample $\bar{x}_2 = \dfrac{\sum_{j=1}^{n_2} x_{2j}}{n_2}$

Variance of the first sample $s_1^2 = \dfrac{\sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2}{n_1}$

Variance of the Second sample $s_2^2 = \dfrac{\sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2}{n_2}$

To test the equality of two means we have to frame the null hypothesis as follows

H_0 : There is no significant difference between the two sample means.

H_0 : The given two samples are drawn from the same population.

Case I : When the population standard deviations σ_1 *and* σ_2 are known

The test statistic can be taken as

$$Z = \frac{t - E(t)}{S.E(t)}$$

$$Z = \frac{(\bar{x}_1 \approx \bar{x}_2) - E(\bar{x}_1 \approx \bar{x}_2)}{SE(\bar{x}_1 \approx \bar{x}_2)}$$

$$|Z| = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right| \approx N(0,1)$$

Case II : When the population standard deviations σ_1 *and* σ_2 are not known

The test statistic can be taken as

$$|Z| = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \right| \approx N(0,1)$$

Case III : When the two samples are drawn from the same population

The test statistic can be taken as

$$|Z| = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} \right| = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \right| \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= Statistic $\pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t)$$

Confidence interval= $\bar{x}_1 \approx \bar{x}_2 \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ [If samples are drawn from different populations]

Confidence interval= $\bar{x}_1 \approx \bar{x}_2 \pm Z_{\frac{\alpha}{2}} \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ [If samples are drawn from Same population]

Confidence interval= $\bar{x}_1 \approx \bar{x}_2 \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ [If population variances are not known]

Test for the significance of single sample proportions :

To test the significance of the single sample proportion, we have to use binomial distribution.

Let $x_1, x_2, x_3, ...x_n$ be a random sample drawn from the Binomial population with the proportion P.

From the given data of sample observations we can calculate the sample proportion $p = \frac{\bar{x}}{n}$

Here $\bar{x}$ = Sample mean n = sample size

Let us consider the population proportion as P and $Q = 1 - P$

To test the significance difference between the sample and population proportions we have to frame the null hypothesis as

H_0 : There is no significance difference between the sample and population proportions

$$i.e., H_0 : p = P$$

Since the size of the sample is large then it follows normal distribution.

To test the above null hypothesis the test statistic Z is given by

$\therefore$ The test statistic $Z = \frac{t - E(t)}{S.E(t)}$

$$Z = \frac{p - E(p)}{SE(p)}$$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= $\text{Statistic} \pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t) = p \pm Z_{\frac{\alpha}{2}} \left(\sqrt{\frac{PQ}{n}} \right)$$

Test for the significance difference between two sample proportions:

Let $x_{11}, x_{12}, x_{13}, \ldots\ldots x_{1n_1}$ and $x_{21}, x_{22}, x_{23}, \ldots\ldots x_{2n_2}$ be the values of two random samples of sizes n_1 and n_2 drawn from binomial population.

From the given data , the two sample proportions are $p_1 = \frac{\bar{x}_1}{n_1}$ $\quad p_2 = \frac{\bar{x}_2}{n_2}$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

To test the equality between sample and population proportions the null hypothesis can be formed as:

H_0 : There is no significance difference between the two sample proportions

$$i.e., \; H_0 : p_1 = p_2$$

Since the samples are large they follow normal distribution

To test the above null hypothesis, the test statistic Z is given by

$$Z = \frac{t - E(t)}{S.E(t)}$$

$$Z = \frac{(p_1 \approx p_2) - E(p_1 \approx p_2)}{S.E(p_1 \approx p_2)}$$

$$|Z| = \left| \frac{p_1 - p_2}{\sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}} \right|$$

$$|Z| = \left| \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \right| \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= $\text{Statistic} \pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t)$$

Confidence interval= $(p_1 \approx p_2) \pm Z_{\frac{\alpha}{2}} \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

Test for the significance of sample standard deviation :

Let $x_1, x_2, x_3, \ldots\ldots x_n$ be a random sample drawn from the normal population with mean μ and standard deviation σ .

From the given data of sample observations we can calculate the sample standard deviation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

To test the equality between the standard deviations of the sample and population the null hypothesis can be framed as follows

H_0 : There is no significance difference between the sample and population standard deviation.

$i.e.,\ H_0 : s = \sigma$

To test the above null hypothesis the test statistic is given by

$$Z = \frac{t - E(t)}{S.E(t)} \quad \Rightarrow Z = \frac{s - E(s)}{S.E(s)}$$

$$Z = \frac{s - \sigma}{\sigma / \sqrt{2n}} \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= $\text{Statistic} \pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t) = s \pm Z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{2n}} \right)$$

Test for the significance difference between the standard deviation:

Let $x_{11}, x_{12}, x_{13}, \ldots\ldots x_{1n_1}$ and $x_{21}, x_{22}, x_{23}, \ldots\ldots x_{2n_2}$ be the values of two random samples of sizes n_1 and n_2 drawn from the two different populations with the parameters μ_1, μ_2 and σ_1^2, σ_2^2.

$i.e.,\ X_1 \approx N(0,1)$ and $X_2 \approx N(0,1)$

From the given two samples

Standard deviation of the first sample $s_1 = \sqrt{\frac{\sum x_{1i}^2}{n_1} - \left(\frac{\sum x_{1i}}{n_1} \right)^2}$

Standard deviation of the Second sample $s_2 = \sqrt{\frac{\sum x_{2j}^2}{n_2} - \left(\frac{\sum x_{2j}}{n_2}\right)^2}$

To test the equality of two standard deviations we have to frame the null hypothesis as follows

H_0 : There is no significant difference between the two sample standard deviations.

H_0 : The given two samples are drawn from the same population.

Case I : When the population standard deviations σ_1 *and* σ_2 are known

The test statistic can be taken as

$$Z = \frac{t - E(t)}{S.E(t)}$$

$$Z = \frac{(s_1 \approx s_2) - E(s_1 \approx s_2)}{S.E(s_1 \approx s_2)}$$

$$|Z| = \left| \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} \right| \approx N(0,1)$$

Case II : When the population standard deviations σ_1 *and* σ_2 are not known

The test statistic can be taken as

$$|Z| = \left| \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} \right| \approx N(0,1)$$

Case III : When the two samples are drawn from the same population

The test statistic can be taken as

$$|Z| = \left| \frac{s_1 - s_2}{\sqrt{\frac{\sigma^2}{2n_1} + \frac{\sigma^2}{2n_2}}} \right| = \left| \frac{s_1 - s_2}{\sigma\sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}}} \right| \approx N(0,1)$$

Here we have to take a critical value at a given specified level of significance. We have to compare the calculated value of Z with the critical value.

If the calculated value is greater than the critical value then we can reject the null hypothesis, otherwise we can accept the null hypothesis at the desired level of significance.

The $100(1-\alpha)\%$ confidence intervals for the population parameter is given by

The $100(1-\alpha)\%$ confidence interval= $\text{Statistic} \pm Z_{\frac{\alpha}{2}} S.E(Statistic)$

$$= t \pm Z_{\frac{\alpha}{2}} S.E(t)$$

Confidence interval= $(s_1 \approx s_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$ [If samples are drawn from different populations]

Confidence interval= $(s_1 \approx s_2) \pm Z_{\frac{\alpha}{2}} \left(\sigma \sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}} \right)$ [If samples are drawn from Same population]

Confidence interval= $(s_1 \approx s_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}$ [If population variances are not known]

Problem 1 : A sample of 900 members with mean and standard deviations are 3.4 and 2.61 respectively, is drawn from a population whose mean is 3.25.

Solution : Given sample size $n = 900$

Sample mean $\bar{x} = 3.4$

Sample standard deviation $s = 2.61$

Population mean $\mu = 3.25$

Here population standard deviation is not known, hence the sample standard deviation is considered as population standard deviation.

$H_0 : \bar{x} = \mu$

(Or)

H_0 : The given sample is drawn from the population whose mean is 3.25

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$Z = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}} = \frac{30(0.15)}{2.61} = \frac{4.5}{2.61}$$

$$Z_{Cal} = 1.7241$$

Here $Z_{Cal} < 1.96$, We can not reject the null hypothesis.

$i.e.,$ The given sample is drawn from the population whose mean is 3.25

Problem 2 : A sample of 100 observations are drawn from a normal population. Mean and variances of the sample are 25 and 10.24 respectively. Is the sample drawn from the population whose mean and variances are 23 and 10.24 respectively ? Also calculate 99% and 95 % confidence interval for the sample mean.

Solution : Given sample size $n = 100$

Sample mean $\bar{x} = 25$

Sample Variance $s^2 = 10.24$

Sample standard deviation $s = \sqrt{10.24} = 3.2$

Population mean $\mu = 23$

Population variance $\sigma^2 = 10.24$

population standard deviation $\sigma = \sqrt{10.24} = 3.2$

$H_0 : \bar{x} = \mu$

(Or)

H_0 : The given sample is drawn from the population whose mean and standard deviations are 23 and 3.2 respectively.

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$Z = \frac{25-23}{\frac{3.2}{\sqrt{100}}} = \frac{10(2)}{3.2} = \frac{20}{3.2}$$

$$Z_{Cal} = 6.25$$

Here $Z_{Cal} > 1.96$, We can reject the null hypothesis.

i.e., The given sample is not drawn from the population whose mean and standard deviations are 23 and 3.2 respectively.

95 % Confidence interval $= t \pm Z_{\frac{\alpha}{2}} \quad S.E(t)$

$$= \bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

$$= 25 \pm 1.96 \frac{3.2}{\sqrt{100}}$$

$$= 25 \pm 1.96 \frac{3.2}{10} = 25 \pm 1.96(0.32) = 25 \pm 0.6272$$

95 % Confidence interval $[25 - 0.6272, \quad 25 + 0.6272] = [24.3728, \quad 25.6272]$

99 % Confidence interval $= t \pm Z_{\frac{\alpha}{2}} \quad S.E(t)$

$$= \bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$

$$= 25 \pm 2.58 \frac{3.2}{\sqrt{100}}$$

$$= 25 \pm 2.58 \frac{3.2}{10} = 25 \pm 2.58(0.32) = 25 \pm 0.8256$$

99 % Confidence interval $[25 - 0.8256, \quad 25 + 0.8256] = [24.1744, \quad 25.0.8256]$

Problem 3 : Mean of a sample of 40 observations is 28. Check whether there is a significant difference between the means of sample and populations whose mean and variances of the population are 27 and 2.56 respectively.

Solution : Given sample size $n = 40$

Sample mean $\bar{x} = 28$

Population mean $\mu = 27$

Population variance $\sigma^2 = 2.56$

population standard deviation $\sigma = \sqrt{2.56} = 1.6$

$H_0 : \bar{x} = \mu$

(Or)

H_0 : There is no significant difference between the sample and population means

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$Z = \frac{28 - 27}{\frac{1.6}{\sqrt{40}}} = \frac{\sqrt{40}\,(1)}{1.6} = \frac{6.3246}{1.6}$$

$$Z_{Cal} = 3.9528$$

Here $Z_{Cal} > 1.96$, We can reject the null hypothesis.

i.e., There is a significant difference between the sample and population means

Problem 4 : Two samples of sizes 40 and 50 are drawn from two different populations whose variances are 3.2 and 2.56 respectively. Means of the two samples are 24.5 and 25 respectively. Check whether there is a significant difference between the two samples.

Solution : Given,

Size of the first sample $n_1 = 40$

Size of the second sample $n_2 = 50$

Variance of the first population $\sigma_1^2 = 3.2$

Variance of the second population $\sigma_2^2 = 2.56$

Mean of the first sample $\bar{x}_1 = 24.5$

Mean of the second sample $\bar{x}_2 = 25$

$H_0 : \bar{x}_1 = \bar{x}_2$

(Or)

H_0 : There is no significant difference between the two sample means.

To test the above null hypothesis, the test statistics is given by

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Z = \frac{24.5 - 25}{\sqrt{\frac{3.2}{40} + \frac{2.56}{50}}}$$

$$Z = \frac{-0.5}{\sqrt{0.1312}}$$

$$Z = -1.3804$$

$$|Z| = 1.3804$$

Here $Z_{Cal} < 1.96$, We can not reject the null hypothesis.

$i.e.,$ There is no significant difference between the two sample means.

Problem 5 : In a random sample of 500 observations mean is found to be 20. In another sample of 400 observations mean is 18.5 Is the two samples drawn from the same population , whose standard deviation is 4.

Solution : Given,

Size of the first sample $n_1 = 500$

Size of the second sample $n_2 = 400$

Mean of the first sample $\bar{x}_1 = 20$

Mean of the second sample $\bar{x}_2 = 18.5$

Population Standard deviation $\sigma = 4$

$H_0 : \bar{x}_1 = \bar{x}_2$

(Or)

H_0 : The given two samples are drawn from the same population whose standard deviation is 4

To test the above null hypothesis, the test statistics is given by

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$Z = \frac{20 - 18.5}{4\sqrt{\frac{1}{500} + \frac{1}{400}}}$$

$$Z = \frac{1.5}{4\sqrt{0.002+0.0025}} = \frac{1.5}{4\sqrt{0.0045}} = \frac{1.5}{0.2683}$$

$$Z = 5.5902$$

Here $Z_{Cal} > 1.96$, We can reject the null hypothesis.

i.e., The given two samples are not drawn from the same population whose standard deviation is 4.

Problem 6 : The average hourly wage of a sample of 150 workers in plant A was Rs. 25.6 with standard deviation of 3.6. The average hourly wage of a sample of 200 workers in plant B was Rs. 28.7 with standard deviation of 3.8. Can an applicant safely assume that the hourly wages paid by plant B are higher than those paid by Plant A.

Solution : Number of workers in plant A is $n_1 = 150$

Number of workers in plant B is $n_2 = 200$

Average hourly wage of workers in plant A is $\bar{x}_1 = 25.6$

Average hourly wage of workers in plant B is $\bar{x}_2 = 28.7$

Standard deviation in plant A is $s_1 = 3.6$

Standard deviation in plant B is $s_2 = 3.8$

H_0 : There is no significant difference between the wages of workers in plant A and plant B

To test the above null hypothesis

$$Z = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \right|$$

$$Z = \left| \frac{25.6 - 28.7}{\sqrt{\frac{(3.6)^2}{150} + \frac{(3.8)^2}{200}}} \right|$$

$$Z = \left| \frac{-3.10}{0.2074} \right|$$

$$Z_{Cal} = 7.7841$$

Here $Z_{Cal} > 1.645$

$\therefore$ We can reject the null hypothesis.

$i.e.,$ There is a significant difference between the wages of workers in plant A and plant B

$\therefore$ Applicant can safely assume that , the hourly wages paid by plant B are higher than the hourly wages of the workers in plant A.

Problem : Random samples drawn from two metro cities A and B gave the following data related to heights of adult males

	City – A	City - B
Mean Height	67.42	67.25
Standard deviation	2.58	2.5
Sample size	1000	1200

i. Is there any significant difference between the means ?

ii. Is there any significant difference between the standard deviations ?

Solution :

Size of the sample collected from City – A is $n_1 = 1000$

Size of the sample collected from City – B is $n_2 = 1200$

Mean height of the adult male in City – A is $\bar{x}_1 = 67.42$

Mean height of the adult male in City – B is $\bar{x}_2 = 67.25$

S.D in heights of adult male from the sample collected from City – A is $s_1 = 2.58$

S.D in heights of adult male from the sample collected from City – B is $s_2 = 2.5$

(i) H_0 : There is no significant difference between the two sample means

$i.e.,\ H_0 : \bar{x}_1 = \bar{x}_2$

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{67.42 - 67.25}{\sqrt{\frac{(2.58)^2}{1000} + \frac{(2.5)^2}{1200}}}$$

$$Z = \frac{0.17}{\sqrt{0.0119}} = 1.5584$$

Here $Z_{Cal} < 1.96$, We can't reject null hypothesis

$i.e.,$ There is no significant difference between the two sample means

(ii) H_0 : There is no significant difference between the two sample standard deviations

$i.e.,\ H_0 : s_1 = s_2$

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{2.58 - 2.5}{\sqrt{\frac{(2.58)^2}{2(1000)} + \frac{(2.5)^2}{2(1200)}}}$$

$$Z = \frac{0.08}{\sqrt{0.0059}} = 1.0415$$

Here $Z_{Cal} < 1.96$, We can't reject null hypothesis

$i.e.,$ There is no significant difference between the two sample standard deviations.

Problem 7 : Two samples of sizes 100 and 200 with the standard deviations 2.58 and 2.5 are drawn from two different populations, whose standard deviations are 2.8 and 2.6 respectively. Is there any significant difference between the two samples.

Solution : Size of the first sample $n_1 = 100$

Size of the second sample $n_2 = 200$

Standard deviation of the first sample $s_1 = 2.58$

Standard deviation of the second sample $s_2 = 2.5$

Standard deviation of the first population $\sigma_1 = 2.8$

Standard deviation of the second population $\sigma_2 = 2.6$

H_0 : There is no significant difference between the two sample standard deviations.

To test the above null hypothesis, the test statistic is given by

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} = \frac{2.58 - 2.5}{\sqrt{\frac{(2.8)^2}{2(100)} + \frac{(2.6)^2}{2(200)}}}$$

$$Z = \frac{0.08}{\sqrt{0.0561}} = 1.3378$$

Here $Z_{Cal} < 1.96$, We can't reject null hypothesis

$i.e.,$ There is no significant difference between the two sample standard deviations.

Problem 8 : In a sample of 1000 people in Karnataka 540 are rice eaters and rest are Wheat eaters. Can we assume both rice and wheat eaters are equally popular in the state of Karnataka at 1 % level of significance.

Solution : Given, sample size $n = 1000$

Number of rice eaters $x = 540$

Proportion of rice eaters $p = \frac{x}{n} = \frac{540}{1000} = 0.54$

H_0 : Both rice and wheat eaters are equally popular in the state of Karnataka.

$i.e.,\ P = 0.5$ then $Q = 1 - P = 1 - 0.5 = 0.5$

To test the above null hypothesis , the test statistic is given by

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$Z = \frac{0.54 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{1000}}}$$

$$Z = \frac{0.04}{0.5\sqrt{0.1}} = 2.5298$$

Here $Z_{Cal} < 2.58$

$i.e.,$ We can't reject null hypothesis.

$\therefore$ Both rice and wheat eaters are equally popular in the state of Karnataka.

Problem 9 : 20 people were attacked by a disease and 1 were survived. Will reject the hypothesis of survival rate , if attacked by the disease is 85 % in favor of the hypothesis at 5 % level.

Solution : Size of the sample $n = 20$

Number of people survived $x = 18$

Proportion of survival in the sample $p = \frac{x}{n} = \frac{18}{20} = 0.9$

$H_0 :\ P = 0.85 \quad H_1 : P > 0.85$ (Right tailed test)

$i.e.,\ P = 0.85$ then $Q = 1 - P = 1 - 0.85 = 0.15$

To test the above null hypothesis , the test statistic is given by

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$Z = \frac{0.9 - 0.85}{\sqrt{\frac{(0.85)(0.15)}{20}}}$$

$$Z = \frac{0.05}{\sqrt{0.0064}} = 0.625$$

Here $Z_{Cal} < 1.645$

i.e., We can't reject null hypothesis.

Problem 10 : Random sample of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favor of the proposal, are same against that they are not at 5 % level of significance.

Solution : Number of men $n_1 = 400$

Number of women $n_2 = 600$

Number of men favor for the proposal to construct the flyover is $x_1 = 200$

Number of women favor for the proposal to construct the flyover is $x_2 = 325$

Proportion of men favor for the proposal to construct the flyover is $p_1 = \frac{x_1}{n_1} = \frac{200}{400} = 0.5$

Proportion of women favor for the proposal to construct the flyover is $p_2 = \frac{x_2}{n_2} = \frac{325}{600} = 0.5417$

H_0 : There is no significant difference between the opinion of men and women about the construction of flyover.

$$\therefore \; P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 325}{400 + 600} = \frac{525}{1000} = 0.525$$

$$Q = 1 - P = 1 - 0.525 = 0.475$$

To test the above null hypothesis, the test statistic is given by

$$|Z| = \left| \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \right|$$

$$|Z| = \left| \frac{0.5 - 0.5417}{\sqrt{(0.525)(0.475)\left(\frac{1}{400} + \frac{1}{600}\right)}} \right|$$

$$|Z| = \left| \frac{-0.0417}{0.0322} \right|$$

$Z_{Cal} = 1.3187$

Here, $Z_{Cal} < 1.96$

i.e., We can't reject the null hypothesis.

$\therefore$ There is no significant difference between the opinion of men and women about the construction of flyover.

Problem 11 : In a survey, 800 persons out of 1000 are found to be tea drinkers before excise duty. After increasing excise duty, 800 persons out of 1200 are found to be tea drinkers. Check whether there is a significant decrease in consumption of tea after the increase of excise duty.

Solution : Size of the sample taken before increase of excise duty $n_1 = 1000$

Number of tea drinkers, before increase of excise duty $x_1 = 800$

Proportion of tea drinkers, before increase of excise duty $p_1 = \frac{x_1}{n_1} = \frac{800}{1000} = 0.8$

Size of the sample taken after increase of excise duty $n_2 = 1200$

Number of tea drinkers, after increase of excise duty $x_2 = 800$

Proportion of tea drinkers, after increase of excise duty $p_2 = \frac{x_2}{n_2} = \frac{800}{1200} = 0.6667$

H_0 : There is no significant decrease in consumption of tea after increase of excise duty

$$\therefore \; P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{800 + 800}{1000 + 1200} = \frac{1600}{2200} = 0.7273$$

$$Q = 1 - P = 1 - 0.7273 = 0.2727$$

To test the above null hypothesis, the test statistic is given by

$$|Z| = \left| \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \right|$$

$$|Z| = \left| \frac{0.8 - 0.6667}{\sqrt{(0.7273)(0.2727)\left(\frac{1}{1000} + \frac{1}{1200}\right)}} \right|$$

$$|Z| = \left| \frac{0.1333}{0.0189} \right|$$

$Z_{Cal} = 7.0529$

Here, $Z_{Cal} > 1.96$

$i.e.,$ We can reject the null hypothesis.

$\therefore$ There is a significant decrease in consumption of tea after increase of excise duty